AMBULANCE OPERATIONS

UNIT 3, PART 2: Prehospital CTAS

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Lecture 2: CTAS Category Definitions and Presentations

CEDIS

- Canadian Emergency Department Information System
- Used to identify the Chief Complaint (or presenting complaint)
- Standardized list of 18 categories (based on body systems)
- Prehospital CTAS uses the same list as ED
- Within the 18 categories there are ~ 150 presenting or chief complaints
  - Presenting complaints have some modifications to better reflect prehospital use (i.e. ENT is one category instead of 3)

CEDIS Categories	
• Cardiovascular • ENT • Environmental • Gastrointestinal • Genitourinary • Mental Health • Neurologic • Obstetrics/Gynecology (OB/GYN)	• Ophthalmology • Orthopedic • Pediatric • Respiratory • Skin • Substance Misuse • Trauma • General and Minor

Organized Approach to Assigning CTAS

- Step 1: **Quick Look**
  - In many cases an accurate CTAS can be applied at first look when presenting complaint is obvious. This is to be used for CTAS 1 designation only

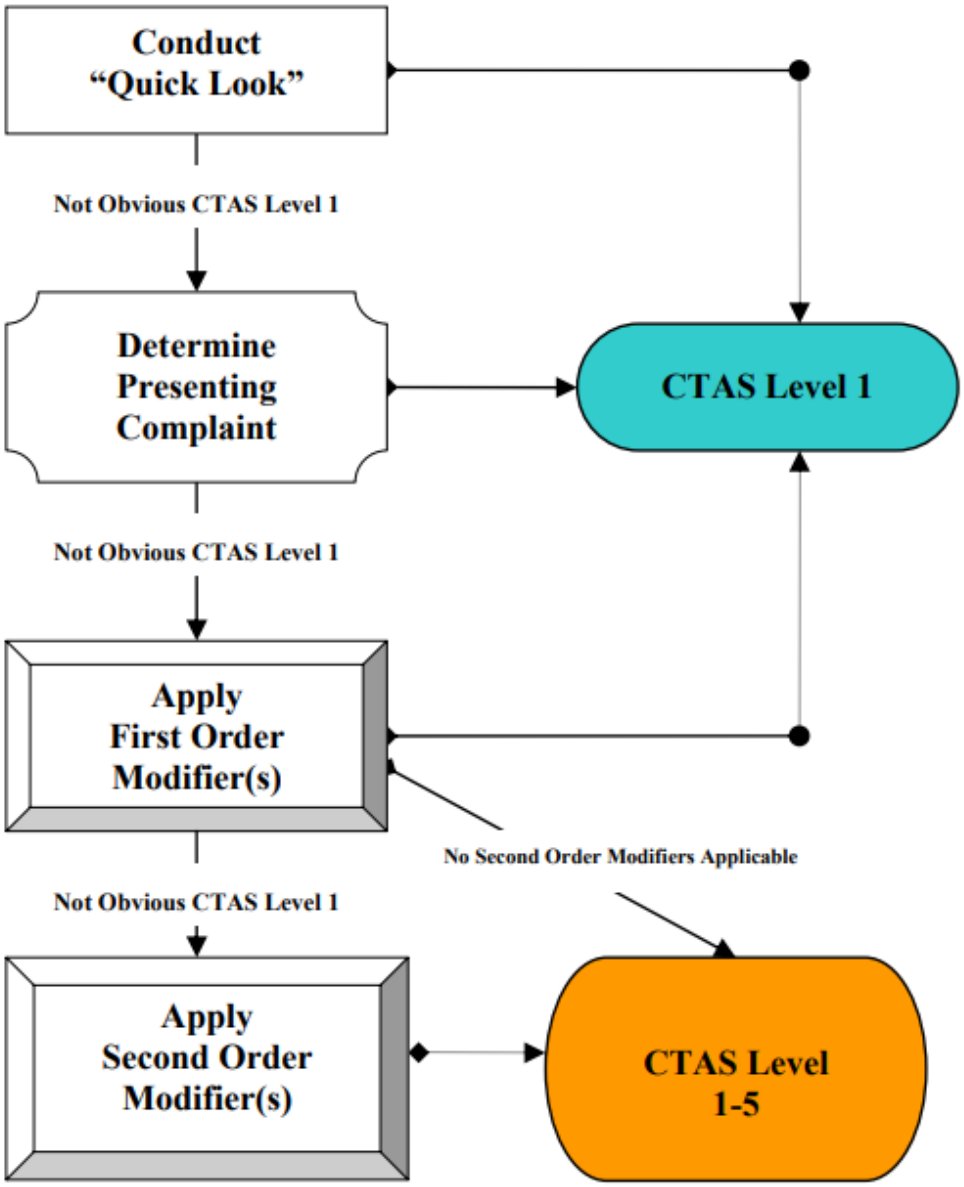
Organized Approach to Assigning CTAS

- Step 2: **Presenting Complaint**
  - 2nd step if not a CTAS 1
  - Based on the CEDIS categories
  - Can be used to assign a CTAS 1
  - If not CTAS 1 → move to step 3

Organized Approach to Assigning CTAS

- Step 3: First and Second Order Modifiers
  - Are used when the “quick look” and “presenting complaint” alone are not enough to assign CTAS
  - Are determined during the primary and /or secondary surveys
  - Are used when the presenting complaint places the patient in more than one CTAS level

Organized Approach to CTAS Determination – Adult



General Principles

- lists are examples and not meant to be all inclusive
- the level is based primarily upon chief complaint
- several chief complaints appear in many acuity levels indicating various severity's to each problem
- The highest CTAS level using all applicable modifiers must be assigned (i.e if pain scale places the patient higher than vital signs, the higher level must be used)

In the ED, the CTAS assessment is used as a ***triage tool*** while in the prehospital care setting it is used solely as an **indicator of acuity**

First Order Modifiers

- Group 1
  - Respiratory distress
  - Hemodynamic Status
  - Level of Consciousness
  - Temperature

- Group 2
  - Pain Score
  - Bleeding disorder
  - MOI

First Order Modifiers

- Group 1: Respiratory
  - CTAS levels are based on
    - level of distress
    - O2 sats

ADULT ***First Order Modifiers, Group One*** **Table 1**
Respiratory Distress

Level of Distress	Oxygen Saturation	CTAS Level
Severe: Fatigue from excessive work of breathing, cyanosis, single-word speech, unable to speak, upper airway obstruction, lethargic or confused	<90%	**1**
Moderate: Increased work of breathing, speaking phrases or clipped sentences, significant or worsening stridor but the airway protected.	<92%	**2**
Mild/Moderate: Dyspnea, tachypnea, shortness of breath on exertion, no obvious increased work of breathing, able to speak in sentences, stridor without any obvious airway obstruction	92-94%	**3**
None	≥94%	**4, 5**

First Order Modifiers

Hemodynamics

- Consider skin colour, temp, moisture and pulse
- We are looking at the perfusion status

Hemodynamic Status	CTAS Level
Shock. Evidence of severe end-organ hypoperfusion: Marked pallor, cool skin, diaphoresis, weak or thready pulse, hypotension, postural syncope, significant tachycardia or bradycardia, ineffective ventilation or oxygenation, decreased level of consciousness. Could also appear as flushed, febrile, toxic, as in septic shock.	1
Hemodynamic Compromise. Evidence of borderline perfusion: pale, history of diaphoresis, unexplained tachycardia, postural hypotension (by history), feeling faint on sitting and standing, or suspected hypotension (lower than normal blood pressure or expected blood pressure for a given patient).	2
Vital signs at the upper and lower ends of normal as they relate to the presenting complaint, especially if they differ from the usual values for the specific patient.	3
Normal Vital signs	4, 5

Signs of Hypoperfusion

- confusion, agitation or decreased LOA,
- mottling and cold extremities,
- marked pallor,
- (decreased O2 saturation on room air).
  - B/P < 90 mmHg systolic (adult)

First Order Modifiers

LOC

- Assessing neurological function
- Looking for responses outside the patient's norm (i.e. dementia patient's, chronic neurological dysfunction)

Status – Level of Consciousness	GCS	CTAS Level
Unconscious: Unable to protect airway, response to pain or loud noise only and without purpose, continuous seizure or progressive deterioration in level of consciousness	3 - 9	1
Altered level of consciousness: Response inappropriate to verbal stimuli, loss of orientation to person, place or time, new impairment of recent memory, altered behaviour	10 - 13	2
Normal: Other modifiers are used to define	14 - 15	3, 4, 5

First Order Modifiers

Temperature

- Defined as ≥ 38.5°C

In the adult pt.

- Based on SIRS to identify early sepsis

Temperature ≥38.5°	CTAS Level
Immunocompromised: neutropenia (or suspected), chemotherapy or immunosuppressive drugs including steroids	2
Looks septic: patient has evidence of infection, have 3 SIRS criteria positive, or show evidence of hemodynamic compromise, moderate respiratory distress or altered level of consciousness	2
Looks unwell: patient has <3 SIRS criteria positive but appear ill-looking (i.e. flushed, lethargic, anxious or agitated)	3
Looks well: patient has fever as their only positive SIRS criteria and appear to be comfortable and in no distress	4

Group 2: Pain as a Marker for Acuity Level

- pain severity is matched to the potential seriousness of the problem to determine acuity level
- Location is used to assess for potential seriousness
  - Central – Head to groin; within a cavity or organ
  - Peripheral – skin, soft tissues, bones
- ***head***, ***chest*** and ***abdominal*** pain (*central*) is of higher acuity than an equal pain involving an extremity (*peripheral*) injury
  - e.g. abdominal pain 8/10 is level 2 but extremity pain 8/10 from a closed # may be a level 3
- similar chronic recurring pain can usually be given a lower level

Note: if peripheral/extremity pain is expected to be caused by a life or limb threatening condition then it should be scored as central pain.

referred pain such as kehrs sign or cardiac jaw pan

*Pain as a Marker for Acuity: Acute versus Chronic

Acute – new onset pain

Chronic – recurring pain

- 32 year old male with chronic back pain complains of same pain but worse than normal at 8/10
  - CTAS 3

versus

- 32 year male fell off the roof and has pain 8/10
  - CTAS 2

*Pain as a Marker for Acuity: Acute versus Recurring Pain

- 42 year old female with severe migraine pain 8/10
  - CTAS 3

versus

- 42 year old female with sudden onset of acute severe headache pain 8/10 with no prior history of severe headaches
  - CTAS 2

Bleeding disorders

- Patients with:
  - known bleeding disorder
  - severe liver failure or
  - taking anticoagulants
- Are at greater risk for hemorrhage
- Require rapid intervention
- This modifier is applied to all trauma and non-traumatic complaints of bleeding.

MOI

- Looks at how energy is transferred during a traumatic event;
- The potential for a serious injury that may not be initially obvious

Mechanism of Injury	CTAS Level 2
General Trauma	**Motor Vehicle Collisions** • Ejection (partial or complete) from vehicle • Rollover • Extrication time ≥20 minutes • Significant intrusion into passenger's space (≥ 0.3 metres occupant site; ≥0.5 metres any site, including roof) • Death in the same passenger compartment • Impact ≥40 km/h (unrestrained) or impact ≥60 km/h (restrained) **Motorcycle Collision** • Impact with a vehicle ≥30 km/h, especially if rider is separated from motorcycle **Fall** • From ≥6 metres (one storey is equal to 3 metres) **Penetrating Injury** • To head, neck, torso or extremities proximal to elbow and knee
Head Trauma	**Motor Vehicle Collision** • Ejection (partial or complete) from vehicle • Unrestrained passenger striking head on windshield **Pedestrian** • Struck by vehicle **Fall** • From ≥1 metre or 5 stairs **Assault** • With blunt object other than fist or feet
Neck Trauma	**Motor Vehicle Collision** • Ejection (partial or complete) from vehicle • Rollover • High speed (especially if driver unrestrained) **Motorcycle Collision** **Fall** • From ≥1 metre or 5 stairs **Axial Load to the Head**

Second Order Modifiers

- These are applied:
  - AFTER the presenting or chief complaint is determined
  - AFTER first order modifiers are applied

- If the patient is obviously a CTAS 1 then continue with the CTAS 1; no need for further justification
- For CTAS 2-5, apply second order modifiers and assign new CTAS level as applicable

General Principles

- experience can be used to “up triage”
- “ never down triage” based on experience
- ***“ if they look sick, then they probably are!”***
- ***remember that there is no absolute right answer- it is based on your assessment and experience***

Vital Signs in Acuity Determination

patients with abnormal vital signs are level 1 or 2

CTAS 1:

- unstable vital signs
- abnormal vital signs with signs of hypoperfusion

CTAS 2:

- abnormal vital signs without signs of hypoperfusion

Vital signs

Blood Glucose Level	Symptoms	CTAS Level
<3 mmol/L	Confusion, seizure, diaphoresis, behavioural change, acute focal deficits	2
	None	3
≥18 mmol/L	Dyspnea, tachypnea, dehydration, thirst, weakness, polyuria	2
	None	3

Blood Pressure	Symptoms	CTAS Level
Systolic Blood Pressure ≥220 or Diastolic Blood Pressure ≥130	**Any** other symptoms (e.g. headache, chest pain, shortness of breath or nausea).	2
Systolic Blood Pressure ≥220 or Diastolic Blood Pressure ≥130	**No** symptoms.	3
Systolic Blood Pressure 200 - 220 or Diastolic Blood Pressure 110 - 130	**Any** other symptoms (e.g. headache, chest pain, shortness of breath or nausea).	3
Systolic Blood Pressure 200 - 220 or Diastolic Blood Pressure 110 - 130	**No** symptoms	4, 5

*Unstable versus Abnormal Vital Signs in Level 1 versus Level 2

- 22 year old female, MVC, rigid painful abdomen, BP 85/60, pulse 130, RR 20
  - CTAS 1

- 72 year old female, moderate SOB, able to speak 4-5 word sentences, BP 85/60, pulse 130 (radial weak but present), RR 20, pale, no diaphoresis
  - CTAS 2

Key Points

- A Change in LOC usually means a CTAS 1 or 2
- pain scales are frequently used in the assignment of an acuity level
- unstable or abnormal vital signs define both level 1 and 2 patients
- when you are not sure, consider other problems in the same acuity level to define your current patient as one of equal severity
- We assign a CTAS 3 times on ACR (contact-departure-arrival)

*Using Time to Physician as a Guide When Uncertain

- after your assessment and response to treatment determine acuity
- base it on chief complaint, vital signs, pain scales etc.
- if unsure then ask yourself how soon after arrival do they need to see the physician

*Using Time to Physician as a Guide When Uncertain

- level 1 - needs physician immediately upon your arrival (i.e they are waiting for you)
- level 2 - needs to see the nurse immediately but there is time for the nurse to do their assessment and the physician should see them within 15 minutes

QUESTIONS